GRADES 6–12 · ENGINEERING DESIGN · OUT-OF-SCHOOL-TIME STEM

We build the builders.

The complete Resilient Futures curriculum — a three-pathway engineering program where students design real New Jersey infrastructure, from a first paper tower in grade 6 to a mentored capstone defended at a public STEM Expo.

3 GRADE-BAND PATHWAYS

6–12 STUDENTS SERVED

5 STANDARDS FRAMEWORKS

40+ ALIGNED STANDARDS

STANDARDS KEY — TAP TO FILTER

NGSS / Science

NJSLS Math

NJSLS ELA

Design & Tech

Career Readiness

HOW TO READ THIS CURRICULUM

Every unit names the standards it teaches

Each week or unit below carries the exact standards it addresses, color-coded by framework and shown in the engineering shorthand teachers and reviewers use. Hover any code for its full description. The master matrix at the end consolidates the complete alignment across all three pathways.

01

GRADES 6–8 · MIDDLE SCHOOL

Bridge Builders

A twelve-week engineering studio built on the proven Hillside–Stevens bridge program. Students run a full engineering design cycle on an authentic Hudson River crossing challenge — defining the problem, modeling and costing designs, weighing risk and sustainability, building and load-testing prototypes, and defending their work at a public showcase.

ESSENTIAL QUESTION How can young engineers design a bridge that is strong, affordable, sustainable, and useful to the community?	NJ ANCHOR CHALLENGE Hudson River crossing · aging bridges · flooding & transportation resilience
STUDENT DELIVERABLES Engineering notebook · scaled & digital model · cost estimate · risk/sustainability matrix · tested prototype · final presentation	

1 WEEK

What is STEM, and how do engineers think?

Program norms and the final-competition overview; how math, science, and engineering work together; the **engineering design process** through a quick paper-tower challenge; students open their engineering notebooks.

MS-ETS1-1 · MP1 · MP2 · SL.PE.7.1 · W.IW.7.2 · 9.4.8.CI.2

2
WEEK

Why do we need sustainable infrastructure?

Global challenges and the UN Sustainable Development Goals; a case study of aging and collapsing bridges; students map how a new Hudson crossing could advance specific SDGs. Quick-write: **why our community needs better bridges**.

MS-ESS3-3 · RI.AA.7.7 · W.IW.7.2 · 9.4.8.CT.1

3
WEEK

What exactly is the Hudson bridge problem?

Teams investigate the geographic and social setting of a Hudson crossing using maps, photos, and demographic data; **define criteria and constraints** (span, clearance, budget, environment, community need) and draft a formal problem statement.

MS-ETS1-1 · 7.RP.A · W.AW.7.1 · SL.PE.7.1 · 9.4.8.IML.3

4
WEEK

How do structure and geometry keep bridges standing?

Bridge types — beam, arch, truss, suspension, cable-stayed — and how form follows function; a mini-lab modeling forces in trusses and supports; students sketch two conceptual designs, labeling geometry and load paths.

MS-ETS1-2 · MS-PS2 · 7.G · 8.G · W.IW.7.2

5
WEEK

How can we design and model our bridges digitally?

Students create a scaled drawing and a **digital model** (Tinkercad, SketchUp, or similar); a mini-lesson on scale factor, ratios, and proportional reasoning; basic quantity calculations from their designs.

MS-ETS1-4 · 7.RP.A · 7.EE.3 · MP4 · 8.2.8.ED

6
WEEK

Is our bridge financially sustainable?

Cost estimating with materials sheets and unit costs (per meter of deck, per tower, per cable); simple payback and life-cycle thinking; students compute **cost per unit of load capacity** as an efficiency metric.

MS-ETS1-3 · 7.RP.A.3 · 7.EE · W.IW.7.2

7
WEEK

How do social and environmental factors shape the “best” design?

Social sustainability and environmental justice — who benefits, who bears the costs; teams complete an **impact matrix** scoring designs on social, economic, and environmental dimensions. Debate: is the lowest-cost bridge always best?

MS-ETS1-1/2 MS-ESS3-3 RI.CT.7.8 W.AW.7.1 SL.ES.7.3 9.4.8.CT.2

8
WEEK

What risks matter, and how do we choose among options?

Risk, probability, and risk management with age-appropriate examples (overloading, storms, budget overruns); teams build a weighted **decision matrix** across cost, strength, aesthetics, sustainability, and risk to compare alternatives.

MS-ETS1-2 MS-ETS1-3 7.SP 7.EE.3 SL.PE.7.1 W.AW.7.1

9
WEEK

How can we turn our design into a physical prototype?

Teams build scaled **prototypes** from final drawings under real constraints — limited materials budget, maximum length, minimum deck width — documenting issues and completing at least one redesign before the final build.

MS-ETS1-2/3/4 7.G · 8.G MP5 8.2.8.ED

10
WEEK

How do we test, measure, and improve our bridges?

A **bridge testing day**: apply increasing load to failure, record maximum load and failure mode, collect deflection and cost data, graph results, and compare teams to identify the best design features across the cohort.

MS-ETS1-3 MS-ETS1-4 7.SP 7.RP · 7.EE W.IW.7.2 SL.II.7.2

11
WEEK

How do engineers communicate and justify their decisions?

Students build a slide deck and short report — problem statement, design description, data and trade-offs, and a final claim for why their design is best — with mini-lessons on argument writing, technical vocabulary, and clear visual design.

MS-ETS1-1–3 W.AW.7.1 W.IW.7.2 W.SE.7.6 SL.PI.7.4 · SL.UM.7.5 9.4.8.IML.3 · TL.3

12
WEEK

Which bridge best meets our community's needs? What did we learn?

The capstone **Bridge Showcase & Testing Competition** before families, teachers, and partners; audiences score designs with a decision matrix; each student writes a reflection on how their thinking as an engineer changed over twelve weeks.

MS-ETS1-1–4 SL.PE.7.1 SL.PI.7.4/7.5 W.RW.7.7 · W.7.10 9.4.8.CI.3 · CT.3

02

GRADES 9–12 · HIGH SCHOOL

Infrastructure Fellows

An applied civil-engineering pre-college studio. Fellows take a real local site — a bridge, intersection, stormwater system, water main, or coastline — through professional practice: site analysis, public datasets, CAD, cost and risk modeling, and a formal technical briefing. A companion Smart Cities studio adds sensors, data, and environmental monitoring.

ESSENTIAL QUESTION How do civil engineers design infrastructure that is safe, durable, affordable, sustainable, equitable, and resilient under real constraints?	NJ ANCHOR CHALLENGE Local bridge · intersection & transportation · stormwater & flooding · water infrastructure · coastal resilience
CORE SKILLS Site analysis · public datasets · surveying · CAD · traffic counts ·	STUDENT DELIVERABLES Design brief · technical drawings · calculations · cost estimate · risk

cost & risk · sensors & dashboards · technical briefing	register · sustainability matrix · 15-minute briefing

1 UNIT

Civil engineering & New Jersey infrastructure

What civil engineers design and why infrastructure matters; mapping infrastructure systems; team roles and charters; a case-study gallery of NJ challenges and project-interest ranking.

HS-ETS1-1 N-Q SL.PE.9-10.1 9.2.12.CAP

2 UNIT

Site analysis & defining the problem

Reading a real site with maps, public datasets, and field observation; specifying quantitative criteria and constraints; writing a defensible engineering problem statement.

HS-ETS1-1 A-CED S-ID W.RW.9-10.7

3 UNIT

Structures, forces & geometry in practice

Load paths, member forces, and material behavior; applying geometry and modeling to evaluate structural options against the site's constraints.

HS-ETS1-2 G-MG MP4

4 UNIT

Digital design & CAD

Translating concepts into scaled technical drawings and CAD models; producing the dimensioned documentation engineers use to communicate and build.

HS-ETS1-2 G-MG 8.2.12.ED

5 UNIT

Cost, constraints & project controls

Quantity take-offs and cost estimating; budgets, schedules, and trade-offs; building a risk register and weighing prioritized criteria the way practicing engineers do.

HS-ETS1-3 N-Q A-CED

Stormwater & water infrastructure

6
UNIT

How communities manage water risk and protect public health; a stormwater walk and drainage sketch; green-infrastructure options and a lead-service-line case study.

HS-ETS1-3 HS-ETS1-4 S-ID W.IW.9-10.2

7
UNIT

Coastal resilience & environmental justice

Lessons from Hurricane Sandy; living-shoreline and protection options; an environmental-justice matrix and a resilience design charrette balancing protection, ecology, cost, and community.

HS-ETS1-3 HS-ETS1-4 RI.AA.9-10 9.4.12.CT

8
UNIT

Technical briefing & public defense

Synthesizing the project into a professional design brief and a 15-minute technical briefing defended before engineers and community partners — the studio capstone.

HS-ETS1-1–4 SL.PI.9-10.4 SL.UM.9-10.5 W.AW.9-10.1

+
STUDIO

Companion studio — Smart Cities, Sensors & Environmental Monitoring ·

GRADES 7–11

How sensors, data, and mapping help communities understand environmental conditions — sensor setup and calibration, data collection on flood-prone grounds, heat islands, air quality, and runoff, with dashboards and ethical data use. Deliverables: monitoring plan, dataset, dashboard, site map, recommendation memo.

HS-ETS1-4 S-ID S-IC 8.1.12.DA 9.4.12.TL

03

GRADES 11–12 · ADVANCED

STEM Research Scholars

A mentored research and design intensive for advanced students. Scholars work in teams with university and professional mentors to investigate a real New Jersey resilience question, run a methodology, analyze data, and produce a capstone defended at a public STEM Expo — a true on-ramp to college research and engineering pathways.

ESSENTIAL QUESTION	MENTORING MODEL
How can student researchers investigate a real resilience challenge and produce evidence-based recommendations?	Near-peer college mentors and professional engineers via the Stevens partnership
STUDENT DELIVERABLES	
Research proposal · literature review · methodology · dataset & analysis · capstone poster & paper · oral defense at the STEM Expo	

1 PHASE

Framing the research question

Identifying a real New Jersey resilience problem; scoping a researchable question with measurable criteria and constraints; team charters and mentor matching.

`HS-ETS1-1` `W.RW.11-12.7` `9.2.12.CAP`

2 PHASE

Literature & precedent review

Locating and evaluating credible sources, prior solutions, and local precedents; synthesizing the state of knowledge into a written review with citations.

`RI.AA.11-12` `W.SE.11-12.6` `9.4.12.IML`

3 PHASE

Methodology & design

Designing a sound method — data collection, modeling, or prototyping — that can actually answer the question; planning for validity, sampling, and ethical data use.

HS-ETS1-2 S-IC N-Q

4
PHASE

Data collection & analysis

Carrying out the method; managing and analyzing data; interpreting results against the criteria and acknowledging uncertainty and limitations.

HS-ETS1-4 S-ID S-IC 8.1.12.DA

5
PHASE

Recommendations & trade-offs

Translating findings into evidence-based recommendations; weighing prioritized criteria, cost, equity, and environmental impact; building the capstone argument.

HS-ETS1-3 W.AW.11-12.1 9.4.12.CT

6
PHASE

Capstone & the public STEM Expo

Producing a research poster and paper and defending the work in an oral presentation at a public STEM Expo before families, partners, and professional engineers.

HS-ETS1-1–4 SL.PI.11-12.4 SL.UM.11-12.5 W.11-12.10

CONSOLIDATED ALIGNMENT

Standards alignment matrix

The complete set of standards addressed across all three pathways, grouped by framework. Middle-grades codes follow the NJSLS 2023 revisions; high-school codes name the alignment targets carried across the Infrastructure Fellows and Research Scholars studios.

NGSS / NJSLS-Science
ENGINEERING DESIGN

MS-ETS1-1 Define criteria and constraints of a design problem.

NJSLS-Mathematics
GRADES 7–8 & HIGH SCHOOL

7.RP.A.1–3 Scale drawings, unit rates, efficiency metrics.

MS-ETS1-2 Evaluate competing design solutions systematically.

MS-ETS1-3 Analyze test data to identify the best characteristics.

MS-ETS1-4 Develop a model for iterative testing and modification.

MS-ESS3-3 Minimize human environmental impact (extension).

MS-PS2 Forces and interactions (extension).

HS-ETS1-1→4 Define, design, evaluate, and model solutions to complex problems.

7.EE.3–4 Multi-step calculations in cost and constraints.

7.G · 8.G Geometry of trusses, scale, similarity.

7.SP · 8.SP Analyze test data; scatter plots and association.

N-Q · A-CED Quantities and modeling equations (HS).

G-MG · S-ID · S-IC Geometric modeling and data analysis (HS).

MP1–MP6 Standards for Mathematical Practice throughout.

NJSLS-English Language Arts

GRADES 7–12

W.AW.1 Argument writing — defend the best design.

W.IW.2 Informative/explanatory technical writing.

W.RW.7 · W.SE.6 Short research; gather and cite evidence.

RI.AA · RI.CT · RL.CT Evaluate arguments on infrastructure and justice.

SL.PE.1 · SL.II.2 Collaborative discussion; interpret data.

SL.PI.4 · SL.UM.5 Present findings with multimedia.

Design, Technology & CS

NJSLS 8.1 & 8.2

8.2.8.ED.1–7 Apply the engineering design process.

8.2.8.ITH · NT Technology's effect on people and the environment.

8.2.8.ETW · EC Ethics, environment, and effects of technology.

8.2.12.ED High-school engineering design and systems.

8.1.12.DA Data and analysis; computational tools.

Career Readiness

NJSLS 9.2 & 9.4

9.2.8.CAP.1–20 Career awareness, exploration, and planning (Gr 8).

9.2.12.CAP Career preparation and planning (HS).

9.4.8/12.CI Creativity and innovation.

9.4.8/12.CT	Critical thinking and problem solving.
9.4.8/12.IML	Information and media literacy.
9.4.8/12.TL	Technology literacy.

OUT-OF-SCHOOL-TIME FIT

How the curriculum serves the six 21st CCLC components

Resilient Futures runs as a STEM-themed program in which the engineering curriculum is the enrichment spine and the other required out-of-school-time components are built around it.

COMPONENT 01

Academic remediation

Daily certified-teacher tutoring in ELA and mathematics, tied to the same NJSLS standards the engineering units apply.

COMPONENT 02

Academic enrichment

The three engineering pathways above — the project-based, standards-aligned core of the program.

COMPONENT 03

Positive youth development

Authentic team roles, leadership through studio captains and the youth advisory board, and near-peer mentoring.

COMPONENT 04

Cultural & arts

Technical drawing, model-making, media documentation, and graphic communication of student work.

COMPONENT 05

Health, nutrition & fitness

Daily movement and wellness blocks plus nutrition education, connected where natural to public-health and built-environment themes.

COMPONENT 06

Parental involvement

Monthly family engagement, family STEM nights, the Bridge Showcase, and the public STEM Expo.

Resilient Futures

A standards-aligned STEM engineering program for grades 6–12, built on the proven Hillside–Stevens engineering model and designed for free, year-round out-of-school-time delivery.

Program partners

Aedifica — program operator

Hillside Public Schools — school collaboration

Stevens Institute of Technology — university partner

Curriculum frameworks

NGSS / NJSLS-Science

NJSLS-Mathematics

NJSLS-English Language Arts

Design, Technology & CS (8.1 / 8.2)

Career Readiness (9.2 / 9.4)

Curriculum derived from the Bridging Brilliance / Building Bridges program and the Resilient Futures studio designs. High-school standard codes indicate alignment targets and should be confirmed against the current NJSLS revisions before publication.